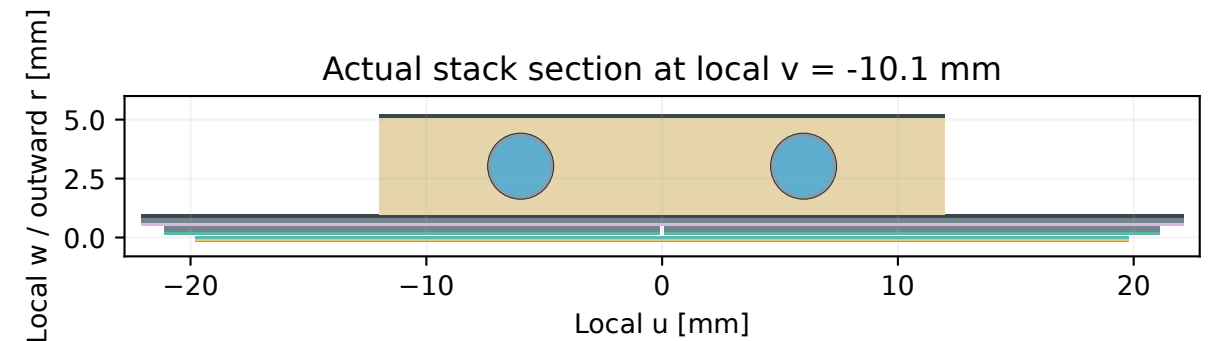
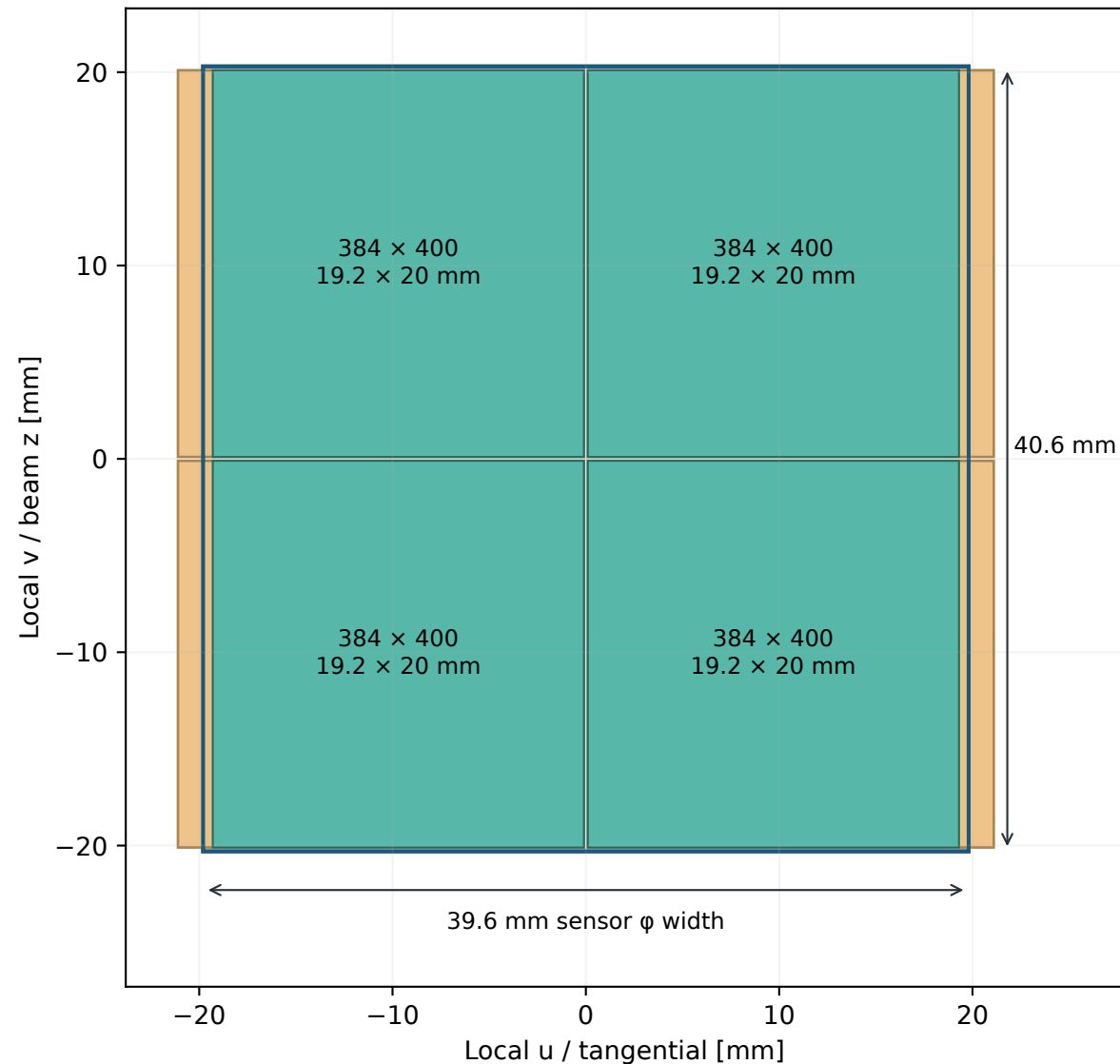


Pixel barrel baseline — quad module

Transverse readout periphery; minimum inactive z edges

Sensor face and underlying ASIC footprints



Selected PR34 quad module · 4 ASICs

Sensor: $39.6 \times 40.6 \times 0.15$ mm

ASIC footprint: $21 \phi \times 20$ z mm

ASIC thickness: 0.15 mm; bump standoff: 0.020 mm

Graphite contact: 0.255 mm

Sensor z guard: 0.200 mm per edge; ϕ guard: 0.500 mm

ASIC peripheral allowance: 1.800 mm on outer ϕ edge(s)

Exposed die ledge beyond sensor: 1.300 mm (approximate model)

Front flex: epoxy / patterned Cu / polyimide / patterned Cu.

Outward support: interface / insulation / graphite / CFRP /

foam+glue with two Ti/CO₂ tubes / CFRP.

Orange: die footprint; green: active matrix; blue outline: sensor.

Bond loops, pads and local pickup connections remain unresolved.

No wire bonding or fastening allowance is placed at the z ends.